Multi-Modal Machine Learning Approach for Threat Detection in WhatsApp and Telegram: Spam Links, Document Based Malware, Chat Behavior, and Sensitive Content Analysis

ML Based sensitive content detection and alert system for WhatsApp and Telegram

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Project Proposal Report

B.Sc. (Hons) in Information Technology Specializing in Cyber Security

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously publish or written by another person expect where the acknowledgement is made in the text.

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Abstract

The rapid growth of messaging platforms like WhatsApp and Telegram has increased the risk of users—especially minors—being exposed to inappropriate or sensitive multimedia content. To address this challenge, this component of the project focuses on developing a lightweight, machine learning (ML)–based detection and alert system for images and videos shared through these platforms. The proposed solution will be integrated into a cross-platform mobile application, functioning as a parental control–like feature.

The system will scan media files before they are sent. When a user selects an image, it will automatically undergo sensitive content detection; for videos, playback and scanning will occur before transmission. Instead of disabling or enabling the send button, the system will employ a floating UI element that overlays on top of the messaging interface. This floating UI will display the detection results—such as sensitivity score, warnings, or risk levels—without interrupting the natural workflow. Users will be able to interact with the floating UI to acknowledge alerts and decide whether to proceed with sending.

The detection engine will utilize optimized deep learning models trained on curated datasets of sensitive and non-sensitive media. It will be designed for mobile performance, ensuring real-time scanning without compromising user experience. The floating UI will provide privacy-aware feedback through tiered alerts such as warnings, soft blocks, or permissions, depending on the confidence level of the detection.

This research contributes by introducing an open-source framework for sensitive content detection in encrypted messaging platforms, an area often dominated by closed commercial tools. The solution emphasizes ethical handling of media, lightweight deployment, and community-driven improvement to enhance online safety.

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List of Abbreviations

ML	-	Machine Learning
AI	-	Artificial Intelligence
NLP	-	Natural Language Processing
CV	-	Computer Vision
API	-	Application Programming Interface
UI	-	User Interface
UX	-	User Experience
TP	-	True Positive
FP	-	False Positive
FN	-	False Negative
TN	-	True Negative
IoT	-	Internet of Things (if needed for integration context)
GPU	-	Graphics Processing Unit
CPU	-	Central Processing Unit
JSON	-	JavaScript Object Notation (common for data exchange in mobile apps)
SDK	-	Software Development Kit
OCR	-	Optical Character Recognition (if text detection is included)
FPS	-	Frames Per Second (video scanning performance)
F1-score	-	Harmonic mean of precision and recall (model evaluation metric)

I. INTRODUCTION

1. Background and Literature Review

The popularity of instant messaging platforms such as WhatsApp and Telegram has transformed communication, particularly among younger users. However, this convenience also exposes users to sensitive and inappropriate content, including nudity, violence, and explicit media, which can negatively affect emotional well-being and lead to digital exploitation. Studies have shown that minors and inexperienced users are particularly vulnerable because of their lack of awareness and the absence of effective parental monitoring tools on these platforms.

Current messaging applications provide limited controls for filtering multimedia content. While some commercial solutions exist, most are closed-source, expensive, or not designed for real-time integration with encrypted platforms. Additionally, many parental control apps operate at a device or network level, often requiring intrusive permissions, which can affect usability and privacy.

Recent advancements in machine learning (ML) and computer vision (CV) provide opportunities to automatically analyze multimedia content before transmission. Pre-trained deep learning models such as convolutional neural networks (CNNs), MobileNet, and EfficientNet have demonstrated high accuracy in image and video classification tasks, including sensitive content detection. Mobile-optimized frameworks now enable lightweight deployment, making real-time analysis on smartphones feasible without significant performance degradation.

Research in this domain focuses on three main areas:

- Sensitive media detection using ML – Models trained on curated datasets (e.g., NSFW image detection, violent content classification).

- Privacy-aware parental controls – Tools that analyze media without storing or sharing user data.

- Platform integration challenges – Working within app restrictions and end-to-end encryption while ensuring user-friendly workflows.

This component of the project addresses these challenges by designing a mobile-sensitive content detection and alert system specifically for WhatsApp and Telegram. The system will scan outgoing media, temporarily block the send button, analyze content, and present a sensitivity score. This design mimics parental control behavior while respecting user privacy.

2. Research Gap

While current research in sensitive content detection has shown strong advances in multimedia classification, there remain significant gaps that limit its application in encrypted messaging platforms such as WhatsApp and Telegram.

Research [1] introduces CNN-based nudity detection models that achieve high accuracy in image classification. However, they are not lightweight enough for mobile devices and do not integrate with real-time messaging platforms or privacy-aware mechanisms.

Research [2] explores sensitive video detection using deep learning, and while it achieves good precision in video content analysis, it lacks real-time encrypted app integration and user-facing features such as floating alerts or soft blocks.

Research [3] provides parental control filtering tools that include blocking mechanisms. However, these rely on cloud-based detection, raising privacy and scalability concerns, and they are not designed for encrypted messaging apps.

Thus, there is a clear gap: no existing solution provides a lightweight, on-device, ML-based sensitive content detection and alert system that is optimized for encrypted messaging apps like WhatsApp and Telegram while maintaining privacy, real-time performance, and community-driven extensibility.

Research Paper	Lightweight ML for Mobile	Real-time Detection on Encrypted Messaging Apps	Tiered Alerts / Soft Blocks	Privacy-aware & Community-driven
Research [1]	Not Implemented	Not Implemented	Implemented (Image classification only)	Not Implemented
Research [2]	Not Implemented	Not Implemented	Implemented (Video content detection)	Not Implemented
Research [3]	Not Implemented	Not Implemented	Implemented (Blocking tools)	Not Implemented
Proposed Solution	Implemented	Implemented	Implemented	Implemented

Table 1 – Research Gap in Sensitive Content Detection

3. Research Problem

Current sensitive content detection mechanisms in messaging platforms are largely dependent on traditional filtering or cloud-based analysis methods. These approaches are often computationally heavy, require continuous internet access, and are not suitable for real-time integration with encrypted applications like WhatsApp and Telegram.

Another important challenge is that most existing parental control and content moderation tools focus only on static image filtering, leaving video content largely unaddressed. Furthermore, they often lack a user-friendly workflow, such as scanning media before sending, and provide limited or no feedback to the user regarding the sensitivity level of the content.

This situation increases the demand for lightweight, on-device, machine learning–based content detection mechanisms that can work seamlessly within resource-constrained mobile environments. Such solutions must ensure real-time detection and alerting while preserving user privacy and supporting multiple media types.

One promising direction is the use of mobile-optimized deep learning models (e.g., MobileNet, EfficientNet) that can classify sensitive images and videos accurately using smaller computational resources. By integrating these models into messaging workflows, users can be warned or prevented from sharing sensitive content without compromising usability or security.

II. OBJECTIVES

1. Main Objective

The main aim of this research is to design and develop a **lightweight, machine learning–based sensitive content detection and alert system** for WhatsApp and Telegram. The system will scan outgoing images and videos in real time, temporarily disable the send function, analyze the content, and provide a risk-level alert to the user, thereby reducing accidental or harmful sharing of explicit media and improving online safety, especially for minors.

2. Sub Objectives

This will be achieved by:

1. Studying existing content detection approaches and identifying their limitations in encrypted messaging platforms, particularly with respect to mobile optimization, video analysis, and privacy concerns.

2. Collecting and curating a dataset of sensitive and non-sensitive images and videos (nudity, violence, explicit content) for training and testing machine learning models.

3. Developing and training lightweight ML models (e.g., MobileNet, EfficientNet) capable of classifying sensitive content efficiently on mobile devices.

4. Designing an in-app scanning workflow that integrates with WhatsApp and Telegram to scan media before sending, with automatic analysis for images and playback analysis for videos.

5. Implementing a risk-based alert mechanism that provides tiered feedback (warning, soft block, allow) based on model confidence levels, ensuring a parental control–like user experience.

6. Optimizing the system for mobile deployment, ensuring minimal resource usage while maintaining high accuracy and user privacy through on-device processing.

7. Testing and validating the system in different scenarios to evaluate performance, accuracy, speed, and usability, ensuring effectiveness in real-world environments.

III. METHODOLOGY

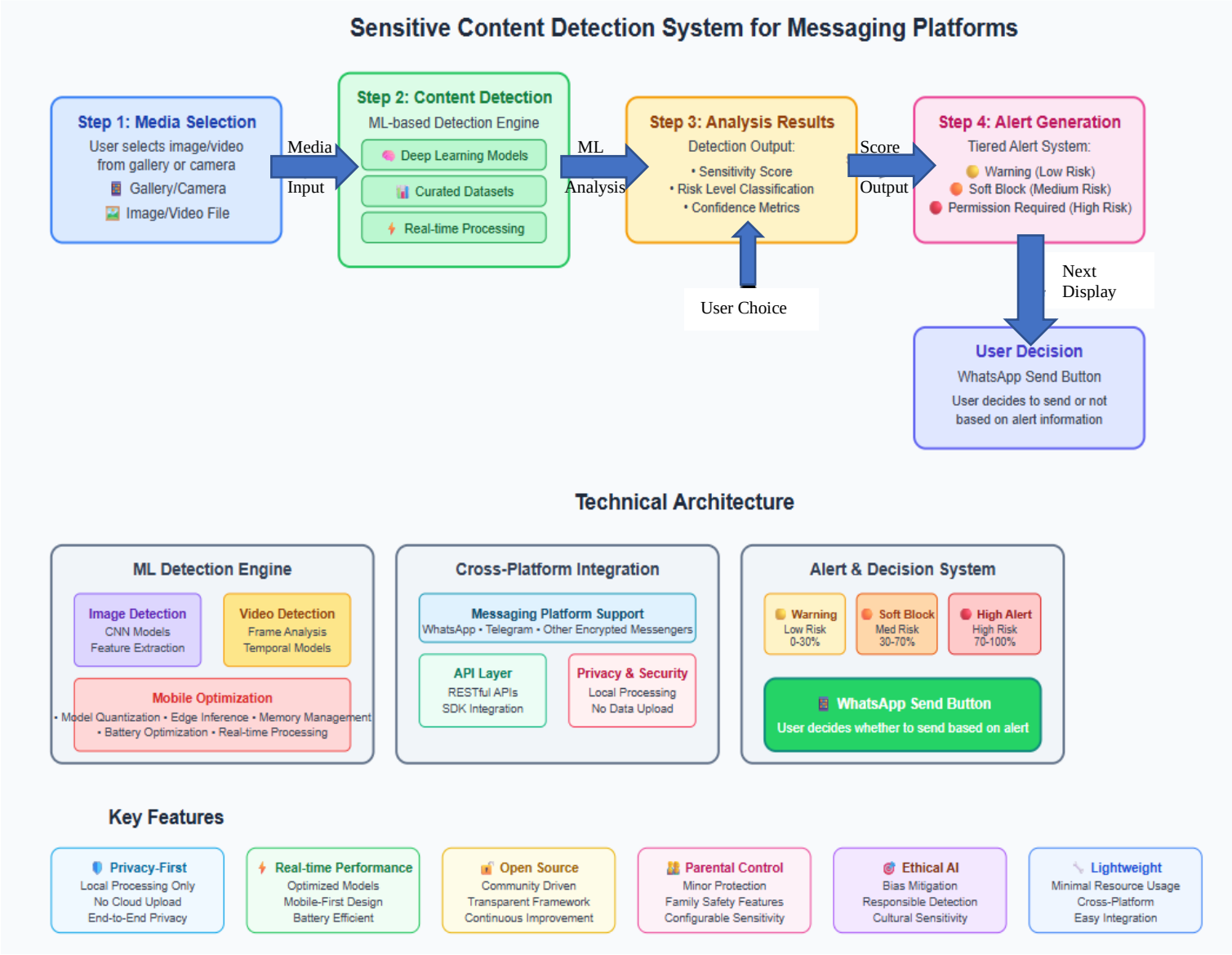

Figure 1- System Diagram

The system diagram illustrates a Sensitive Content Detection System for Messaging Platforms, designed to work with apps like WhatsApp and Telegram. The process begins with media selection (image or video from gallery/camera), followed by on-device ML-based content detection using deep learning models and curated datasets. The system generates analysis results such as sensitivity scores, risk classifications, and confidence metrics. Based on this, a tiered alert system (warning) guides the user, who then makes the final decision via the send button. The technical architecture highlights mobile-optimized ML detection, cross-platform integration, and a privacy-first design with local processing. Key features include real-time performance, lightweight models, parental control, ethical AI, and strong privacy protection.

1. Requirement Gathering

In the requirement gathering process, a combination of literature review, consultation with professionals, and analysis of technical and user requirements was carried out. This process ensured that the proposed solution is feasible, efficient, and addresses real-world challenges in messaging platforms like WhatsApp and Telegram.

- Extensive Literature Review: A review of the latest research and journal articles on machine learning–based image and video classification, parental control systems, and content filtering in encrypted messaging platforms was conducted. This provided insights into existing approaches, their limitations (such as lack of mobile optimization and privacy issues), and potential areas for improvement such as pre-send scanning and tiered alerts.

- Stakeholder Consultation: Discussions were carried out with cybersecurity experts, mobile app developers, and child online safety professionals. These consultations helped to validate the need for real-time sensitive content detection, parental control–like mechanisms, and privacy-preserving solutions. User surveys and feedback sessions further highlighted the importance of integrating a non-intrusive and user-friendly alert system.

- System Requirements Analysis: Technical requirements were identified, including lightweight ML model integration, support for both images and videos, pre-send scanning workflows, and risk-level classification. Performance requirements such as low latency, high accuracy, and efficient mobile resource utilization were also noted. Security and privacy requirements emphasized on-device processing to avoid storing or transmitting sensitive user data.

- Use Case Scenarios: Multiple use case models were developed to demonstrate how the system operates in real-world settings. For instance:

    - A child attempting to share an explicit image via WhatsApp, where the send button is temporarily blocked until a sensitivity alert is shown.

    - A video containing violent scenes being scanned during playback before transmission on Telegram.

    - A user receiving tiered alerts (warning, soft-block, allow) based on model confidence levels.

These scenarios ensure that the proposed solution addresses both functional and practical aspects of sensitive content detection in messaging environments.

2. Past Research Analysis

The analysis of prior research provides valuable insights into existing solutions for sensitive content detection and highlights clear gaps that limit safe, private, real-time deployment on encrypted messaging platforms (WhatsApp, Telegram). Below is a concise synthesis of three representative studies and what they contribute - and omit - relative to the goals of a lightweight, on-device, privacy-aware detection & alert system.

A. Literature Review of Key Studies

Research [1] - CNN-based image nudity detection: Demonstrates high accuracy for explicit-image filtering using convolutional neural networks. Strengths: solid image classification performance and well-tuned CNN architectures for nudity/explicitness. Limitations: models are not optimized for mobile or constrained devices, no discussion of pre-send scanning in messaging apps, and no user interaction/alert design (e.g., tiered warnings or soft blocks).

Research [2] - Deep learning video content detection: Presents frameworks for automatically detecting sensitive scenes in video (nudity, violence, explicit content) with good precision and temporal modelling. Strengths: handles temporal dynamics and per-frame/segment scoring. Limitations: the work is evaluated as an offline or cloud pipeline and is not integrated into real-time communication flows; it does not consider UX elements (disabling send until scan completes, tiered alerts) or privacy constraints of end-to-end encrypted messaging.

Research [3] - Parental control / app-level filtering tools: Describes systems and tools that block or filter inappropriate content across apps, often providing parental control dashboards and policy settings. Strengths: practical deployment focus and user control mechanisms. Limitations: relies on cloud processing which raises privacy and connectivity concerns, lacks lightweight on-device ML models and does not adapt to encrypted messaging workflows where server-side inspection is infeasible.

B. Identification of Gaps

- Lightweight on-device ML for mobile: Research [1] shows strong image models but does not address model compression, pruning, quantization, or architecture choices (e.g., MobileNet/efficient transformers) needed to run in real time on phones.

- On-device video scanning & temporal efficiency: Research [2] handles video well in a server setting but does not present methods for efficient pre-send video scanning on mobile (frame sampling, fast segment scoring, approximate inference).

- Integration with encrypted messaging workflows: None of the three works demonstrates integration into end-to-end encrypted platforms (pre-send scanning hooks, send-button disabling, or client-side enforcement) without breaking privacy.

- Tiered alerts and human-centric UX: Only Research [3] partially addresses blocking but lacks fine-grained, tiered alerting (warnings, soft blocks, permission prompts) informed by model confidence and user context.

- Privacy-preserving operation / minimal data exposure: Research [3] depends on cloud processing; Research [1] & [2] assume data availability for training/testing but do not offer privacy-preserving inference designs (e.g., fully client-side inference, federated learning, or on-device differential privacy).

- Community-driven, open frameworks & ethical handling: None provide an open-source, community-extensible framework with clear ethical guidelines for handling flagged media, dataset curation, and bias/false positive mitigation tailored to messaging platforms.

- Real-time UX tradeoffs & user control: There is little empirical work showing acceptable latency/UX tradeoffs (scan time vs. send delay) or policies that let guardians/older users customize sensitivity thresholds without excessive blocking.

3. Feasibility Study

The technical feasibility of the proposed research incorporates the assessment of whether the tools, technologies, and techniques required in the development and implementation of the ML-based sensitive content detection and alert system are appropriate. This shall be addressed in the following areas:

A. Tools and Technology

- TensorFlow Lite / PyTorch Mobile: TensorFlow Lite and PyTorch Mobile are optimized deep learning frameworks that enable the deployment of machine learning models directly on mobile devices. These frameworks will be employed to implement lightweight CNNs and transformer-based models for detecting sensitive content in images and videos while ensuring efficiency on resource-constrained smartphones.

- Android Studio & Kotlin/Java: Android Studio will serve as the primary development environment for building the mobile application interface and integrating the detection engine into WhatsApp and Telegram workflows. Kotlin/Java will be used to implement

core mobile functionalities, such as scanning triggers, UI for alerts, and integration with pre-send workflows.

- Python: Python will be used for model training, dataset preprocessing, and performance evaluation. Libraries such as OpenCV, TensorFlow, and Scikit-learn will assist in developing and validating the ML models before deployment to mobile platforms.

- SQLite / Local Storage: A lightweight local database (SQLite) or secure storage mechanisms will be used to store sensitivity logs, user settings, and parental control preferences while ensuring data privacy and no leakage outside the device.

- GitHub (Open-Source Framework): GitHub will act as the collaborative open-source platform for sharing code, models, and updates with the research community, supporting community-driven improvement and transparency.

B. Algorithm and Architectures

- Convolutional Neural Networks (CNNs): CNN-based architectures (e.g., MobileNetV3, EfficientNet-Lite) will be employed for detecting nudity, violence, and explicit content in images due to their efficiency in extracting spatial patterns and suitability for mobile devices. [4]

- Temporal Deep Learning Models: For videos, temporal CNNs or lightweight transformer-based models (e.g., MobileViT, Temporal CNN, or Distilled Vision Transformers) will be used to detect sensitive sequences by analyzing multiple frames and aggregating predictions. [5]

- Transfer Learning & Model Optimization: Pre-trained models will be fine-tuned using curated datasets of sensitive and non-sensitive media. Techniques such as pruning, quantization, and knowledge distillation will be used to reduce model size and improve inference speed on mobile devices. [6]

- Risk Scoring Mechanism: The system will assign a sensitivity score (low, medium, high) to each piece of content. Based on this score, the system will determine whether to show a warning, issue a soft block, or allow the user to proceed. [7]

- Alert Mechanism Architecture: The alert system will be integrated into the app interface, temporarily disabling the "send" button until scanning completes and providing tiered alerts (warning, soft block, override options).

C. Techniques

- On-device Inference: The ML models will run fully on the device to ensure privacy-preserving detection without exposing user media to cloud servers, aligning with encrypted messaging principles.

- Performance Evaluation: The system will be evaluated on accuracy, false positive/negative rates, inference speed, and resource usage (CPU, memory, battery) to ensure acceptable tradeoffs for real-time usage. [8]

- Dataset Preprocessing & Augmentation: Sensitive and non-sensitive media datasets will undergo preprocessing (resizing, normalization) and augmentation (rotation, noise addition, blurring) to improve model generalization.

- Real-time Video Frame Sampling: Instead of scanning all video frames, key frames will be sampled at intervals and analyzed to reduce latency while maintaining reliable detection accuracy. [9]

- User Interaction Testing: Usability testing will measure user response to tiered alerts, system transparency, and the acceptability of short delays during scanning.

- Privacy & Ethical Handling: Sensitive media will never be stored or uploaded to external servers. Ethical dataset usage and community-driven review mechanisms will ensure fairness and accountability in model training and deployment. [10]

4. Project Requirements

A. Functional Requirements

- Sensitive Content Detection (Images & Videos): The system must automatically scan all selected images and videos before they are sent via WhatsApp or Telegram. Sensitive content such as nudity, violence, or explicit media should be flagged using lightweight ML models optimized for mobile deployment.

- Real-time Alert and Blocking: The system should temporarily disable the "send" button until the scan is completed. Based on the sensitivity score, the system must generate tiered alerts such as warnings, soft blocks (requiring confirmation), or hard blocks (preventing sending altogether).

- Risk Scoring Mechanism: Each piece of media must be evaluated and assigned a sensitivity score (low, medium, high). The risk score determines the alert level presented to the user, ensuring flexibility and adaptability of the system in different contexts.

- Privacy Preservation: The detection process must occur entirely on the user's device without uploading media to external servers, maintaining compliance with end-to-end encryption principles of WhatsApp and Telegram.

- Parental Control Features: The system should allow guardians to set thresholds for sensitive detection and enforce stricter blocking rules for minors, ensuring customizable content control.

- Logging & Reporting: The system must provide optional local logging of detection results and blocked media attempts. These logs should remain secure and only accessible to the device owner/guardian.

B. Non-Functional Requirements

- Accuracy: The ML models should achieve high detection accuracy while minimizing false positives (blocking safe content) and false negatives (missing harmful content). Accuracy must be validated against diverse datasets of sensitive and non-sensitive media.

- Performance: The system must operate in real time, ensuring minimal latency during media scanning. For example, image scanning should complete within seconds, and video scanning should rely on efficient frame sampling to maintain usability.

- Resource Efficiency: The detection system should consume minimal device resources

(CPU, GPU, memory, and battery). Optimized lightweight models (quantization, pruning) should ensure smooth operation on low- and mid-range smartphones.

- Availability & Reliability: The system must remain stable and functional under continuous use, even with multiple media files being scanned simultaneously. It should not crash or significantly slow down messaging applications.

- Security: The system must ensure that sensitive content remains private on the user's device and is not shared externally. Logs, risk scores, and alerts should be stored securely using encryption, preventing unauthorized access.

- Scalability & Extensibility: The framework should be open-source and designed for extensibility, allowing the community to add new detection categories (e.g., hate symbols, self-harm content) and improve models over time.

- Usability: Alerts and warnings should be clear, user-friendly, and provide simple options (proceed, cancel, override). The system should not overburden users with technical complexity, ensuring smooth integration with everyday messaging use.

5. Work Breakdown & Grant Chart

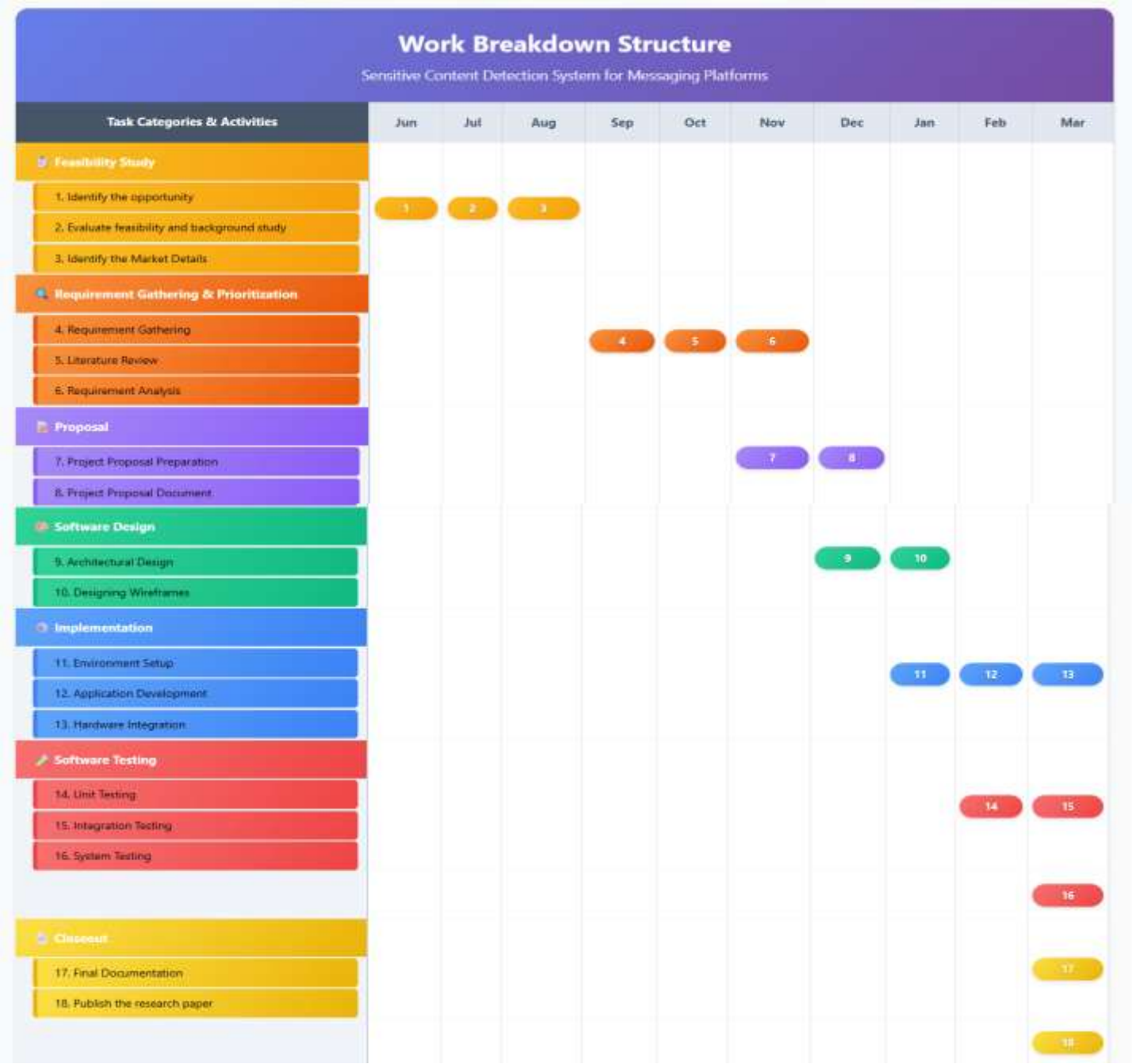

Figure 2 – Grant chart

Figure 3 - Work Breakdown Chart

6. Commercialization

The proposed Sensitive Content Detection System has significant commercialization potential in the rapidly expanding markets of digital safety, parental control technologies, and content moderation tools. With growing concerns around child safety, exposure to inappropriate content, and digital wellbeing, this solution addresses a pressing global demand. The system incorporates lightweight machine learning models optimized for mobile devices, ensuring fast, efficient, and privacy-conscious detection of sensitive images and videos in real time.

The commercialization strategy will focus on parents, educators, app developers, and digital safety organizations, who require reliable tools to monitor and restrict access to sensitive content. Additionally, the solution can be positioned for messaging platforms (e.g., WhatsApp, Telegram), social media apps, and device manufacturers as an integrated feature.

A major advantage of the system lies in its risk-scoring and alert mechanisms (low/medium/high sensitivity), combined with seamless mobile integration and privacy-preserving design. This makes it highly adaptable for diverse user groups while being easy to use and trusted by parents and guardians.

Key Commercialization Points:

- Target Market:
  - Parents and guardians seeking parental control solutions
  - Educational institutions for safe digital learning environments
  - Messaging/social media app developers integrating safety features
  - Smartphone manufacturers offering pre-installed safety tools

- Unique Selling Proposition (USP): Unlike conventional parental control apps, this system combines real-time sensitive content detection, on-device machine learning, and customizable alert levels in one lightweight mobile solution, ensuring speed, accuracy, and user privacy.

- Revenue Model:
  - Subscription-based model for parents and institutions (monthly/annual plans)
  - Licensing agreements with messaging/social media platforms for integration
  - Partnerships with device manufacturers to embed the solution into smartphones/tablets
  - Freemium model, offering basic detection for free with premium advanced features (risk scoring, reporting dashboards)

- Market Entry Strategy: Market entry can begin with pilot programs in schools and parental communities, followed by partnerships with messaging app providers to showcase its effectiveness. Demonstrating compliance with child safety regulations (COPPA, GDPR-K, etc.) will also strengthen adoption. Strategic collaborations with NGOs, child safety advocacy groups, and educational technology providers will provide credibility and faster uptake.

Altogether, given the increasing demand for digital safety tools and the novel combination of lightweight AI, privacy-focused design, and parental control features, this system has a high likelihood of successful commercialization across global markets.

IV. BUDGET AND BUDGET JUSTIFICATION

Since the proposed Sensitive Content Detection System is software-oriented, no specialized hardware components are required for its implementation. The primary costs will be associated with cloud computing resources for training lightweight ML models, dataset collection/curation, and dissemination of results through publications. Additional minor costs are anticipated for project execution such as internet access, software licensing, and stationery.

The detailed budget is presented in **Table** below.

Type	Cost (LKR)
Internet and Web Browsing (research, dataset downloads, testing)	5,000
Cloud Computing / GPU Services (model training, preprocessing)	20,000
Dataset Acquisition & Annotation (curation, labeling sensitive content)	12,000
Software Tools & Licenses (Android Studio plugins, ML libraries if premium)	7,500
Publication Cost (journal/conference submission fees)	15,000
Stationery & Documentation (reports, printing, presentations)	5,500
Total	**65,000**

Table 2 - Budget Analysis

Summary

This project proposal aims to enhance digital safety and parental control by developing a lightweight sensitive content detection system for mobile applications. The focus is on addressing the rising concerns of children's exposure to inappropriate images and videos in messaging and social media platforms. The proposed system leverages optimized machine learning models (e.g., MobileNet/EfficientNet) for on-device detection, ensuring efficiency, accuracy, and user privacy. The project will involve a comprehensive literature review, requirement analysis, dataset collection, and ML model development, followed by integration into a mobile environment with a risk scoring and alert mechanism. The solution will be validated through unit, integration, and usability testing to ensure reliability and effectiveness. Additionally, a commercialization strategy will be explored, targeting parents, educational institutions, and digital platforms to provide a scalable and user-friendly tool for safer digital interactions.

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